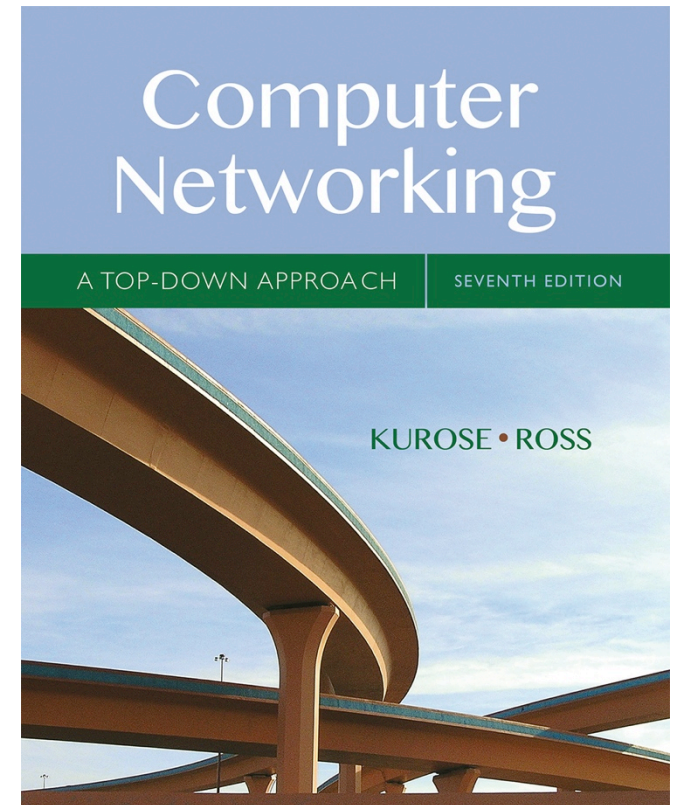


Chapter I

Introduction

©



Computer Networking: A Top Down Approach

7th edition

Jim Kurose, Keith Ross

Pearson/Addison Wesley

April 2016

Chapter 1: roadmap

1.1 *what is the Internet?*

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, network structure

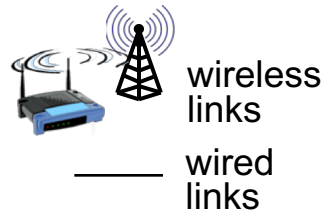
1.5 protocol layers, service models

What's the Internet: a “nuts and bolts” view

components



- billions of connected computing devices:
 - *hosts* = *end systems*
 - running *network apps*

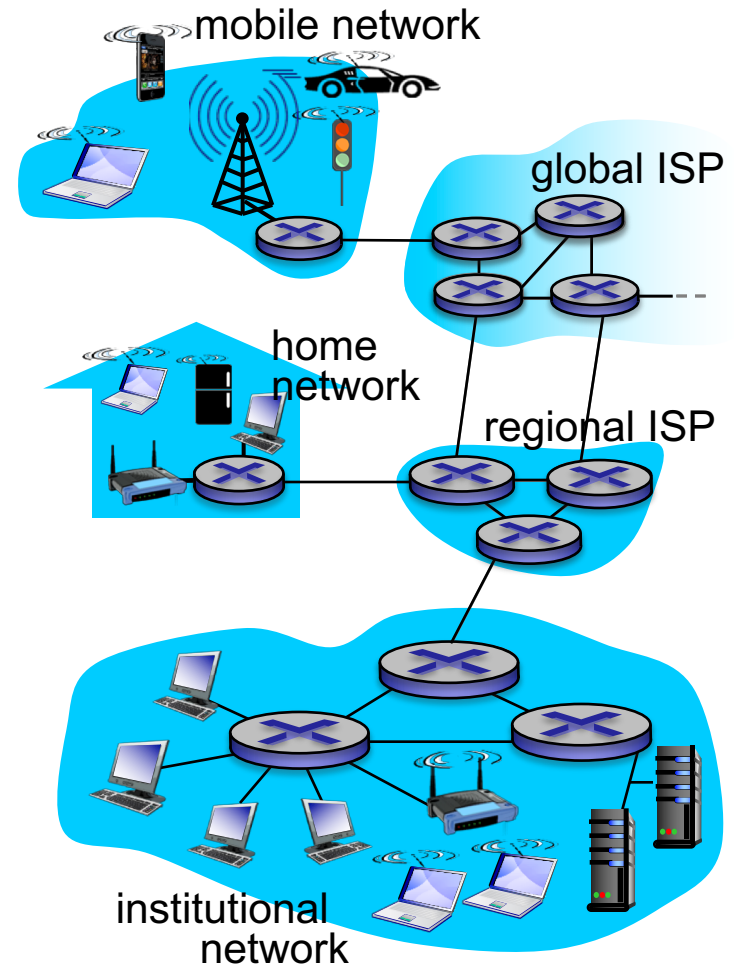


- *communication links*
 - fiber, copper, radio, satellite
 - transmission rate: *bandwidth*



- *switches*: forward chunks of data (packets)
 - *routers* and *switches*

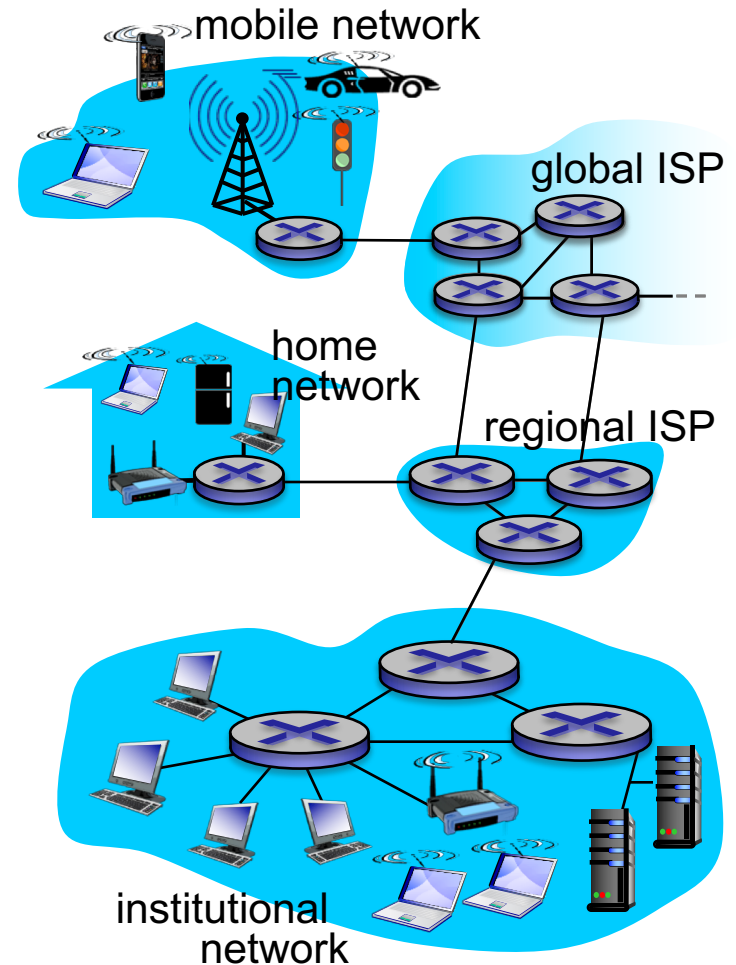
configurations



What's the Internet: a “nuts and bolts” view

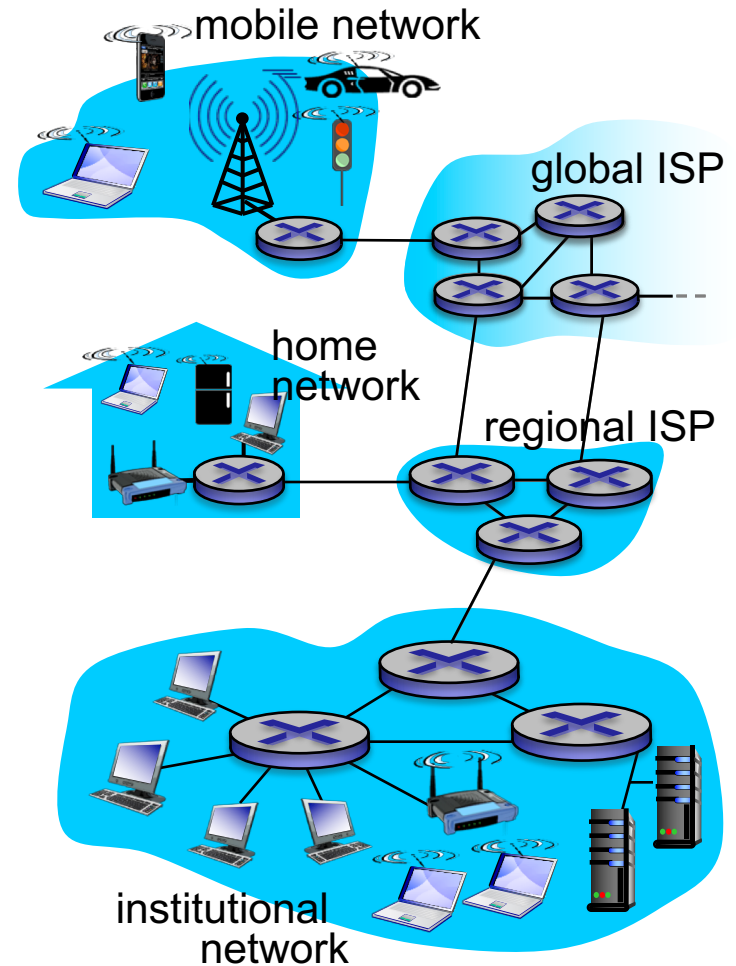
tying it all together

- **Internet “network of networks”**
 - **inter**connected **net**works transport data
- **protocols** control sending, receiving of information – data exchange between end devices
 - e.g., TCP, IP, HTTP, Skype, 802.11
- **Internet standards**
 - IETF: Internet Engineering Task Force
 - RFC: Request for comments
 - IEEE: Inst. of Electrical & Electronic Eng.
 - ITU: Intl Telecomm. Union (UN)



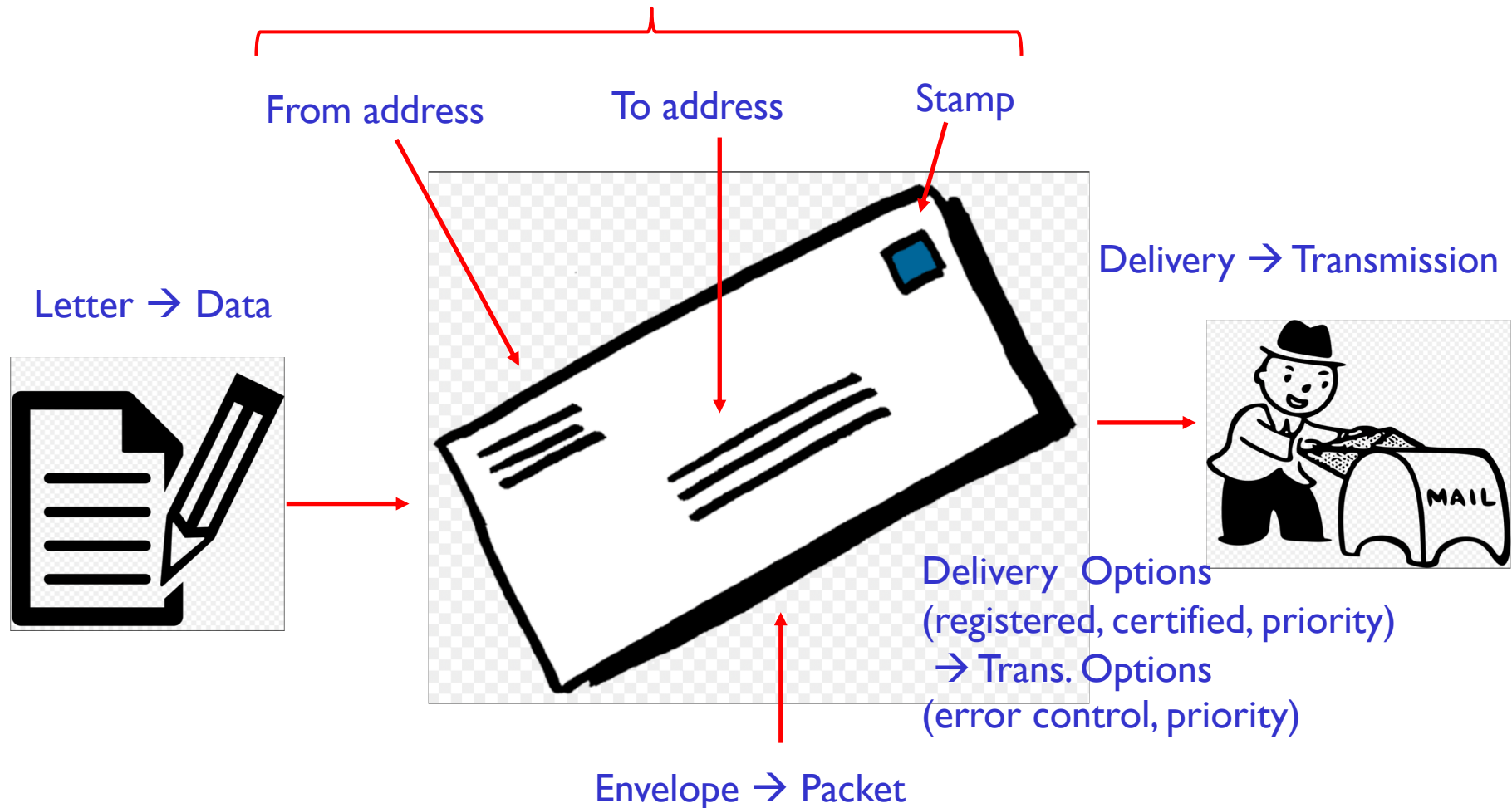
What's the Internet: a "service view"

- *an infrastructure that provides services to **applications**:*
 - Web, VoIP, email, games, e-commerce, social nets, ...
- *it provides **programming interface** to apps*
 - **software modules** (socket) that allow sending and receiving application programs to "connect" to Internet
 - **service options** (options/choices) necessary for the functionality of applications



Postal Service Analogy

Postal Service Rules → Internet Rules



What's a protocol?

human protocols:

- asking a question
- in a classroom
- introductions
- a courtroom
- dinner conversation

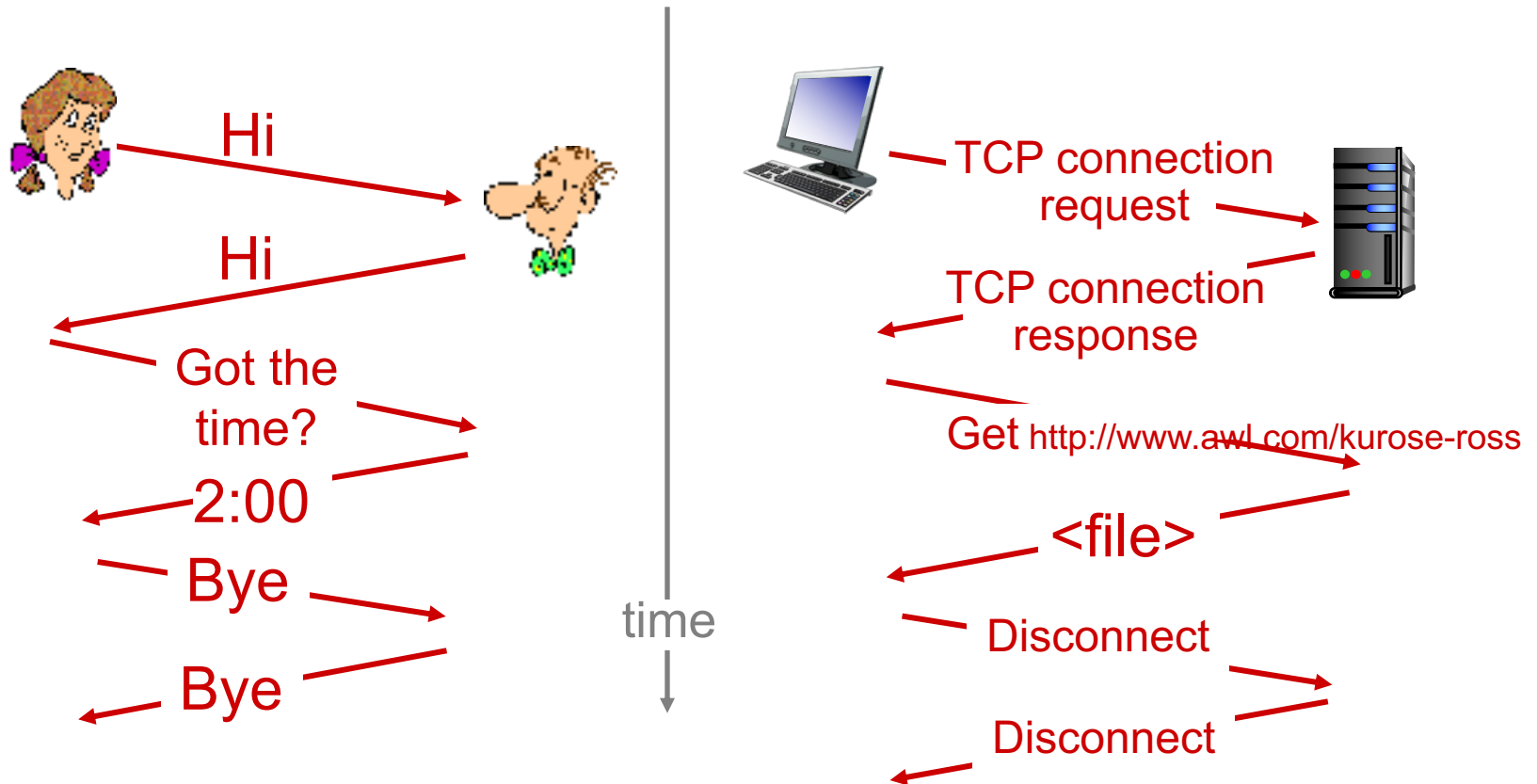
network protocols – similar but:

- devices rather than humans
- diversity of devices
- type of application:
 - queries
 - reports/files

*protocols define format, order of
messages sent and received
and actions taken on
message transmission, receipt*

What's a protocol?

a human protocol and a computer network protocol:



Chapter 1: roadmap

1.1 what is the Internet?

1.2 network edge

- end systems, access networks, links

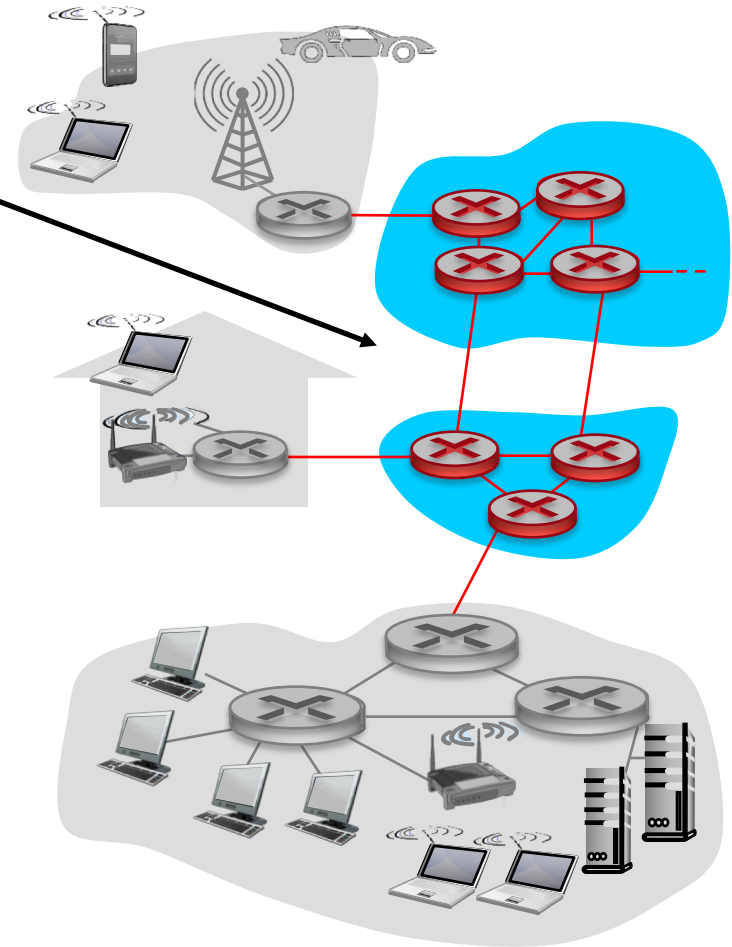
1.3 network core

- packet switching, network structure

1.5 protocol layers, service models

The network core

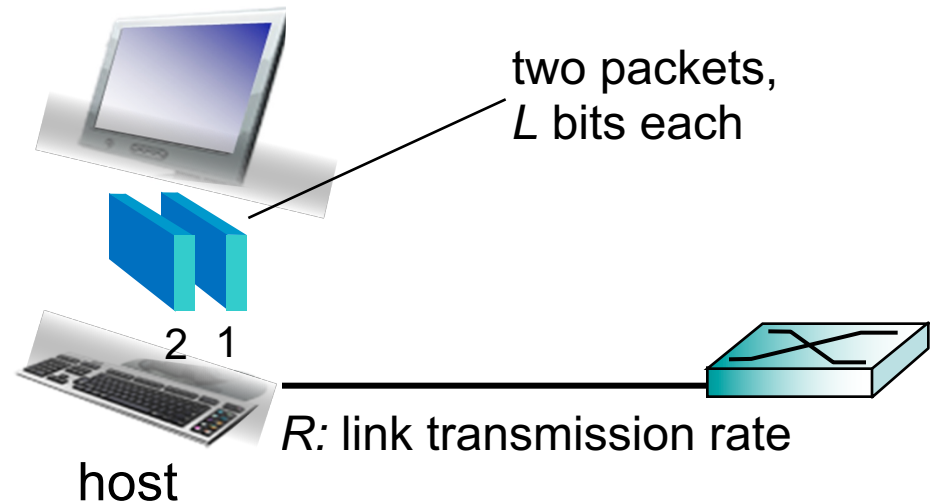
- mesh of interconnected switches referred to as *routers*
- packet-switching: hosts break application-layer *data* into chunks and put into *packets*
 - forward packets from one *router* to the next, across links on path from source to destination
 - each packet transmitted at full link capacity



End devices: generate data

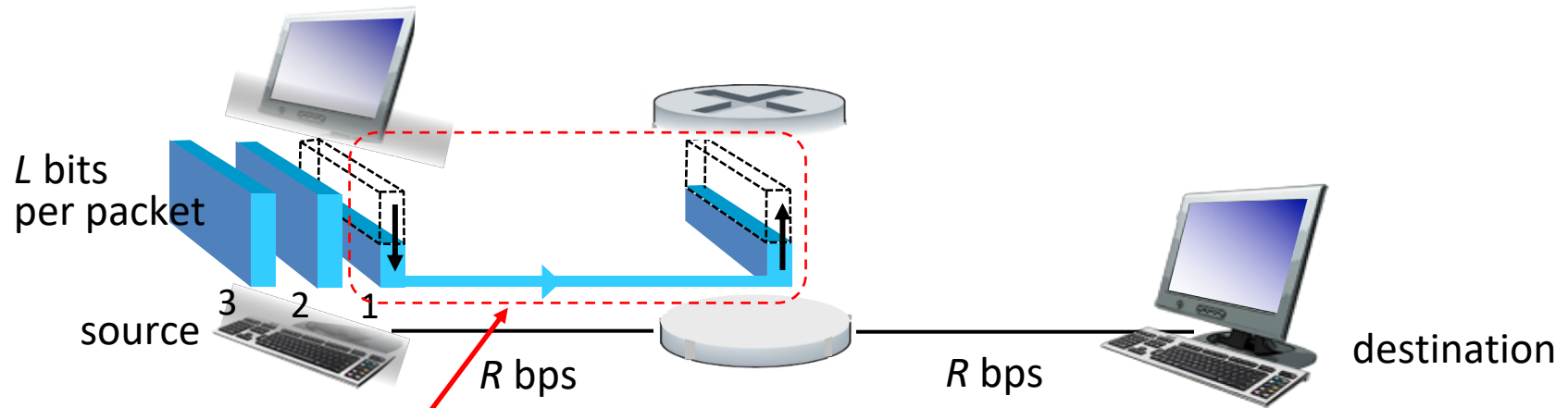
host sending function:

- takes application data
- breaks into smaller chunks, known as *packets*, of length L bits
- transmits packet into access network at *transmission rate R*
 - link transmission rate, aka *link capacity, aka link bandwidth*



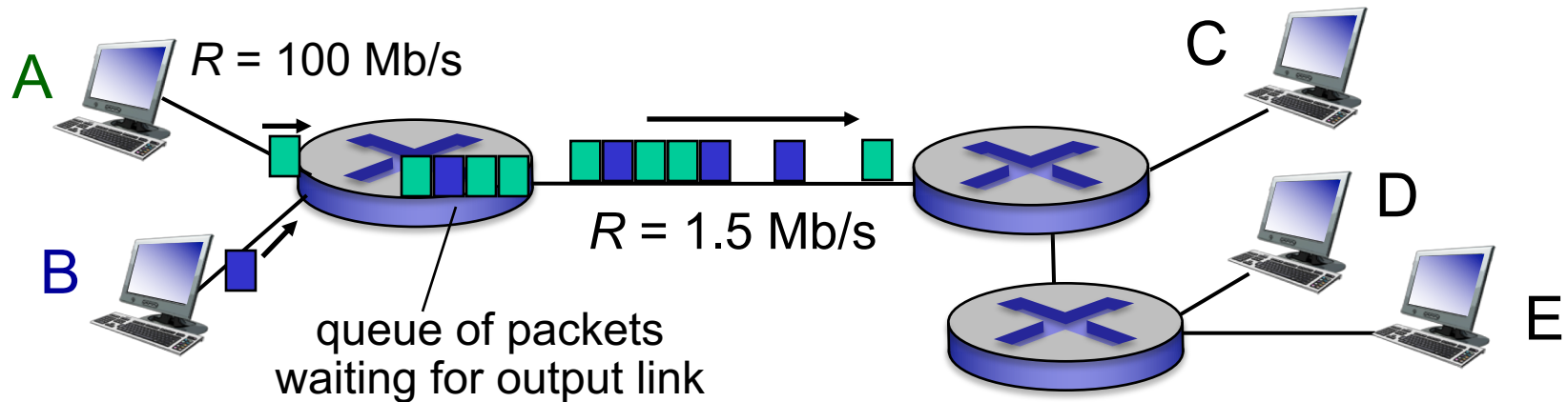
$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet over link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Packet-switching: store-and-forward



- *store and forward*: entire packet must arrive at router before it can be transmitted on next link – *hop by hop*

Packet Switching: queueing delay, loss



queuing and loss:

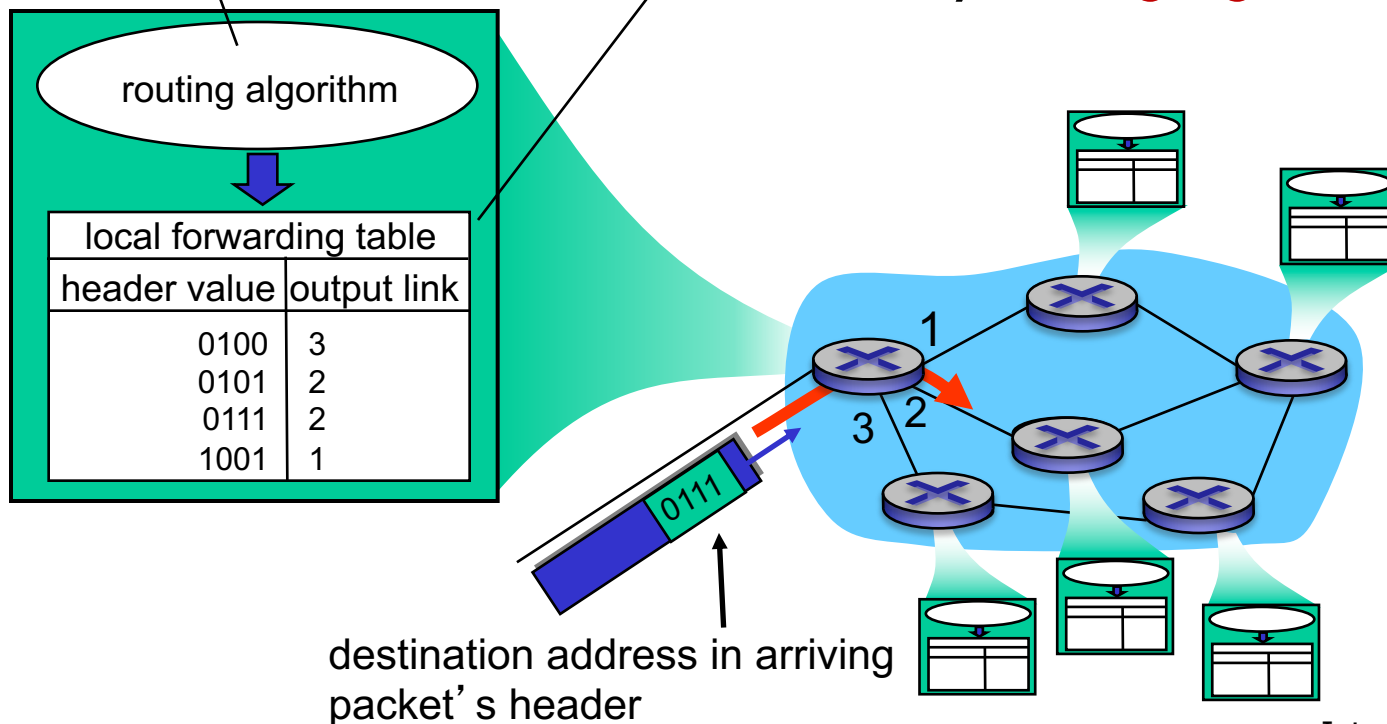
- if arrival rate (in bits) to link exceeds transmission rate of link for a period of time:
 - packets will queue, wait to be transmitted on link
 - packets can be dropped (lost) if memory (buffer) fills up

Two key network-core functions

routing: determines **source-destination route** taken by packets

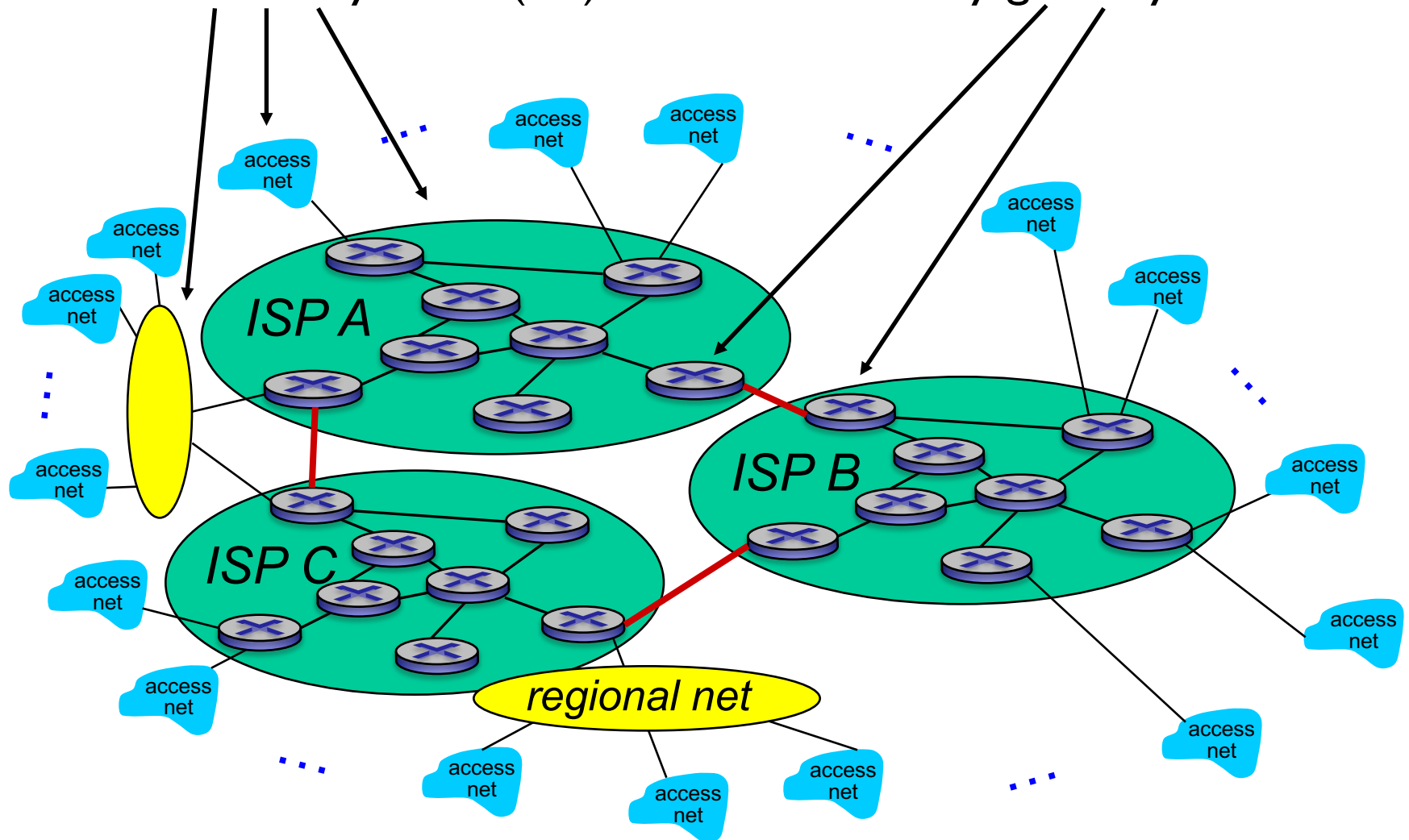
- *routing algorithms*

forwarding: move packets from router's input (port) to appropriate router output (port) based on **forwarding table** created by **routing algorithm**



Internet structure: network of networks

Autonomous Systems (AS) interconnected by gateway routers



Chapter 1: roadmap

1.1 *what is the Internet?*

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, network structure

1.5 protocol layers, service models

Protocol “layers”

*Networks are complex,
with many “pieces”:*

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question:

*how should we organize
such a complex system?*

.... or at least our
discussion of networks?

Analogy: Organization of air travel



- a series of steps